

# AI-Powered Customer Support Automation Platform

Capstone Project — Summer School '26 (n8n)

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## 1. Problem Analysis

### Business Context

A growing software company receives customer support requests through multiple channels — email, website forms, and chat. As the customer base scales, manually categorizing tickets, assigning them to the right agents, tracking resolution, and following up with customers becomes slow and error-prone. This leads to delayed responses, inconsistent prioritization, duplicate effort, and declining customer satisfaction.

This project automates the full support ticket lifecycle — from intake to resolution to feedback — using a modular, AI-powered n8n platform.

### Stakeholders

| Stakeholder        | Interest                                                                   |
|--------------------|----------------------------------------------------------------------------|
| Customers          | Fast acknowledgment, timely resolution, clear communication                |
| Support Agents     | Clear ticket ownership, less manual triage                                 |
| Support Managers   | Visibility into critical issues, workload distribution, escalation control |
| Company Leadership | Customer satisfaction metrics, operational efficiency, cost reduction      |

### Pain Points

- Manual ticket categorization is slow and inconsistent across agents
- No automatic prioritization — critical issues can sit unnoticed alongside minor ones
- No structured escalation path for urgent issues
- Customers aren't proactively updated on ticket status
- No systematic feedback collection or reporting on support performance

### Objectives

- Automatically capture and log every incoming ticket
- Use AI to classify ticket category and priority without human intervention
- Route tickets to the correct team automatically
- Escalate critical (P1) tickets to a human for immediate review and approval
- Keep customers informed at every stage (acknowledgment → assignment → resolution)
- Collect structured feedback after resolution
- Maintain a full audit trail and generate analytics for management

## 2. Workflow Architecture

### Overall System Architecture

The platform consists of 5 independent, interconnected n8n workflows, each responsible for one stage of the ticket lifecycle. Workflows communicate using n8n's Execute Sub-workflow trigger/action pattern, passing structured JSON data forward at each stage. A Google Sheet acts as the central ticket database (Support Tickets DB), with a second tab (Feedback) for survey responses.

### Workflow Interaction Diagram

See the accompanying architecture diagram (02\_Architecture\_Diagram) for the full visual flow. Summary of stages:

```
Customer → [WF1: Ticket Collection & Registration]
           → [WF2: AI Classification & Prioritization] (Gemini)
           → [WF3: Agent Assignment & Escalation]      (Switch + IF)
           → [WF4: Customer Communication & Resolution] (Webhook: resolve)
           → [WF5: Feedback Collection & Analytics]    (Webhook: feedback, Cron: report)
```

### Event Flow Between Workflows

- Webhook (POST /ticket-intake) → Workflow 1 starts execution
- Workflow 1 completes ticket registration → calls Workflow 2 via Execute Sub-workflow, passing ticket fields
- Workflow 2 calls Gemini AI to classify category/priority → updates the sheet → calls Workflow 3
- Workflow 3 routes the ticket (Switch node on category) → checks priority (IF node) → escalates via email if P1 → calls Workflow 4
- Workflow 4 sends the assignment update email immediately. Later, an independent webhook (POST /ticket-resolve) — representing a real agent action — triggers the resolution branch, which calls Workflow 5
- Workflow 5 sends a feedback survey. A separate webhook (POST /submit-feedback) logs customer responses. A Cron-scheduled branch independently reads all tickets and emails a summary analytics report

## 3. Implementation Summary

| Workflow                                        | Node Count (approx.) | Trigger Type                                             |
|-------------------------------------------------|----------------------|----------------------------------------------------------|
| 1. Ticket Collection & Registration             | 5                    | Webhook                                                  |
| 2. AI Ticket Classification & Prioritization    | 4                    | Execute Sub-workflow Trigger                             |
| 3. Agent Assignment & Escalation                | 11                   | Execute Sub-workflow Trigger                             |
| 4. Customer Communication & Resolution Tracking | 7                    | Execute Sub-workflow Trigger + Webhook                   |
| 5. Feedback Collection & Support Analytics      | 8                    | Execute Sub-workflow Trigger + Webhook + Schedule (Cron) |

**Total: ~35 nodes across 5 workflows — exceeding the minimum requirement of 4–6 workflows and 20–25 nodes.**

## 4. Advanced Features Implemented

| Feature                     | Where Implemented                                                                                                                                                                                    |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AI-powered decision making  | Workflow 2 — Google Gemini classifies ticket category and priority from natural language                                                                                                             |
| Human approval step         | Workflow 3 — P1 (critical) tickets trigger an escalation email requiring manager review before proceeding                                                                                            |
| Error handling              | Resolved data-loss issues across Gmail response nodes; used explicit node references ( <code>\$( 'NodeName' ).item.json</code> ) to ensure reliable data flow past nodes that overwrite payload data |
| Logging and audit trail     | Every stage of the ticket lifecycle is logged to Google Sheets (status, category, priority, assigned_team, timestamps)                                                                               |
| Scheduled workflow (Cron)   | Workflow 5 — periodic analytics report generation and email                                                                                                                                          |
| Webhook-triggered workflows | Workflow 1 (ticket intake), Workflow 4 (resolution update), Workflow 5 (feedback submission)                                                                                                         |
| Conditional branching       | Workflow 3 — Switch node (category routing) + IF node (priority-based escalation)                                                                                                                    |

## 5. Tech Stack

- Automation platform: n8n (self-hosted via Docker, WSL2 / Ubuntu 24.04)
- AI model: Google Gemini (via Google AI Studio API)
- Database: Google Sheets (Support Tickets DB — Sheet1 + Feedback tab)
- Notifications: Gmail (OAuth2)
- Trigger types: Webhook (REST), Execute Sub-workflow, Schedule (Cron)